

Challenge (Habanero)

- Q1.** A gamer scored 12 points in 4 minutes. What is the unit rate of points per minute?
(A) 2 points/minute
(B) 3 points/minute
(C) 4 points/minute
(D) 6 points/minute
(E) 8 points/minute
- Q2.** A coding task requires 15 lines of code to process 3000 data points. How many lines of code are required to process 18000 data points?
(A) 60 lines
(B) 75 lines
(C) 90 lines
(D) 120 lines
(E) 150 lines
- Q3.** The score of a game is directly proportional to the number of players. If 4 players scored 120 points, how many points would 6 players score?
(A) 150 points
(B) 180 points
(C) 200 points
(D) 220 points
(E) 240 points
- Q4.** A video game uses 250 MB of data per hour of gameplay. How many hours of gameplay are possible with 3750 MB of data?
(A) 10 hours
(B) 12 hours
(C) 15 hours
(D) 18 hours
(E) 20 hours
- Q5.** A gamer's score increases at a rate of 5 points per level. If the gamer's current score is 50 points, what is the gamer's level?
(A) Level 5
- (B) Level 6
(C) Level 8
(D) Level 10
(E) Level 12
- Q6.** The time it takes to complete a coding task is inversely proportional to the number of coders. If 5 coders take 2 hours to complete the task, how long would it take for 10 coders to complete the task?
(A) 0.5 hours
(B) 1 hour
(C) 1.5 hours
(D) 2 hours
(E) 4 hours
- Q7.** A gamer's data usage is 50 MB per game. If the gamer plays 8 games, what is the total data usage in GB?
(A) 0.2 GB
(B) 0.4 GB
(C) 0.5 GB
(D) 0.8 GB
(E) 1 GB
- Q8.** The cost of a gaming console is directly proportional to the number of controllers. If a console with 2 controllers costs 200, what is the cost of a console with 4 controllers?
(A) 300
(B) 350
(C) 400
(D) 450
(E) 500

Open Ended Questions (FRQ)

- Q9.** A coding task requires x lines of code to process y data points. If 12 lines of code are required to process 600 data points, find the number of lines of code required to process 2400 data points. Show your work and express your answer as a numerical value.
- Q10.** The score of a game is inversely proportional to the time taken to complete the game. If a gamer scores 80 points in 10 minutes, find the score if the gamer takes 20 minutes to complete the game. Show your work and express your answer as a numerical value.

Chef's Tips

- Chef's Tips: Ensure students show all work and use proper mathematical notation.
- Chef's Tips: Encourage students to read the questions carefully and choose the correct units.
- Chef's Tips: Remind students to check their calculations for reasonableness.